

# AUTOM-03: Monaco Configuration-as-Code

**Series:** AUTOM — Dynatrace Automation | **Notebook:** 3 of 8 | **Created:** January 2026 | **Last Updated:** 04/25/2026

Monaco (Monitoring as Code) is Dynatrace's official CLI tool for configuration management. It uses YAML files to define configurations and supports version control, CI/CD integration, and multi-environment deployments.

## Table of Contents

- 1. [Introduction](#)
- 2. [Getting Started](#)
- 3. [Project Structure](#)
- 4. [Configuration Files](#)
- 5. [Common Operations](#)
- 6. [Advanced Features](#)

## Prerequisites

Before starting this notebook, ensure you have:

| Requirement | Description                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------|
| Monaco CLI  | Installed via GitHub releases or brew                                                                                    |
| API Token   | Token with <code>settings.read</code> , <code>settings.write</code> , <code>ReadConfig</code> , <code>WriteConfig</code> |

| Requirement | Description                    |
|-------------|--------------------------------|
| Tenant URL  | Your Dynatrace SaaS tenant URL |

## Learning Objectives

By the end of this notebook, you will:

- Understand Monaco's configuration-as-code model
- Know how to download existing configurations
- Be able to create and deploy configurations via YAML
- Handle multi-environment deployments

## 1. Introduction

### Why Monaco?

| Benefit                  | Description                                 |
|--------------------------|---------------------------------------------|
| <b>Version Control</b>   | Track all changes in Git                    |
| <b>Repeatability</b>     | Deploy same config to multiple environments |
| <b>Review Process</b>    | Use pull requests for config changes        |
| <b>Disaster Recovery</b> | Quickly restore configurations              |
| <b>Documentation</b>     | YAML files serve as living documentation    |

### Monaco vs Direct API

| Aspect           | Monaco             | Settings API    |
|------------------|--------------------|-----------------|
| Format           | YAML files         | JSON payloads   |
| State management | Implicit (by name) | Manual tracking |

| Aspect         | Monaco                             | Settings API     |
|----------------|------------------------------------|------------------|
| Multi-tenant   | Built-in support                   | Custom scripting |
| Dry run        | Yes ( <code>--dry-run</code> )     | No               |
| Delete orphans | Yes ( <code>delete</code> command) | Manual           |

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## 2. Getting Started

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### Installation

#### macOS (Homebrew):

```
brew install dynatrace-oss/monaco/monaco
```

#### Linux/Windows:

```
# Download from GitHub releases
curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/latest/download/monaco-linux-amd64 -o monaco
chmod +x monaco
```

#### Verify installation:

```
monaco version
```

### Environment Setup

Create environment variables for authentication:

```
export DT_TENANT_URL="https://{tenant-id}.live.dynatrace.com"
export DT_API_TOKEN="&lt;your-api-token&gt;"
```

---

## Your First Download

Download all configurations from your tenant:

```
monaco download \
  --environment-url "$DT_TENANT_URL" \
  --api-token "$DT_API_TOKEN" \
  --output-folder ./downloaded-config
```

Download specific configuration types:

```
monaco download \
  --environment-url "$DT_TENANT_URL" \
  --api-token "$DT_API_TOKEN" \
  --output-folder ./downloaded-config \
  --api builtin:management-zones,builtin:tags.auto-tagging
```

---

## 3. Project Structure

### Recommended Layout

```
dynatrace-config/
├─ manifest.yaml           # Project manifest
├─ environments/
│   ├─ development.yaml    # Dev environment config
│   ├─ staging.yaml        # Staging config
│   └─ production.yaml     # Production config
└─ projects/
    └─ my-project/
        ├─ management-zones/
        │   └─ config.yaml
        ├─ auto-tagging/
        │   └─ config.yaml
        └─ alerting-profiles/
            └─ config.yaml
```

### Manifest File

The `manifest.yaml` defines your project:

```
manifestVersion: 1.0

projects:
  - name: my-project
    path: projects/my-project

environmentGroups:
  - name: default
    environments:
      - name: development
        url:
          type: environment
          value: DT_DEV_URL
        auth:
          token:
            type: environment
            value: DT_DEV_TOKEN
      - name: production
        url:
          type: environment
          value: DT_PROD_URL
        auth:
          token:
            type: environment
            value: DT_PROD_TOKEN
```

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## 4. Configuration Files

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### Basic Configuration Structure

```
configs:
  - id: production-zone
    config:
      name: Production
      template: management-zone.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment
```

## Management Zone Example

**config.yaml:**

```
configs:
  - id: production-mz
    config:
      name: Production
      template: production-mz.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment

  - id: staging-mz
    config:
      name: Staging
      template: staging-mz.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment
```

**production-mz.json:**

```

{
  "name": "{{ .name }}",
  "rules": [
    {
      "type": "SERVICE",
      "enabled": true,
      "conditions": [
        {
          "key": {
            "type": "STATIC",
            "attribute": "SERVICE_TAGS"
          },
          "comparisonInfo": {
            "type": "TAG",
            "operator": "EQUALS",
            "value": {
              "context": "CONTEXTLESS",
              "key": "environment",
              "value": "production"
            },
            "negate": false
          }
        }
      ]
    }
  ]
}

```

---

## Auto-Tagging Example

**config.yaml:**

```

configs:
- id: application-tag
  config:
    name: Application
    template: application-tag.json
  type:
    settings:
      schema: builtin:tags.auto-tagging
      scope: environment

```

**application-tag.json:**

```
{
  "name": "{{ .name }}",
  "rules": [
    {
      "type": "SERVICE",
      "enabled": true,
      "valueFormat": "{Service:DetectedName}",
      "propagationTypes": ["SERVICE_TO_PROCESS_GROUP_LIKE"],
      "conditions": []
    }
  ]
}
```

## Using Variables

Variables allow environment-specific values:

**config.yaml:**

```
configs:
- id: alerting-profile
  config:
    name: "{{ .Env.ENVIRONMENT_NAME }} Alerting"
    template: alerting.json
    parameters:
      severity_level: "{{ .Env.ALERT_SEVERITY }}"
  type:
    settings:
      schema: builtin:alerting.profile
      scope: environment
```

## 5. Common Operations

### Deploy Configurations

Deploy to all environments:

```
monaco deploy manifest.yaml
```

Deploy to specific environment:

```
monaco deploy manifest.yaml --environment production
```

Dry run (validate without applying):

```
monaco deploy manifest.yaml --dry-run
```

## Validate Configurations

Check YAML syntax and schema validity:

```
monaco validate manifest.yaml
```

## Delete Configurations

Remove configurations not in your project (orphans):

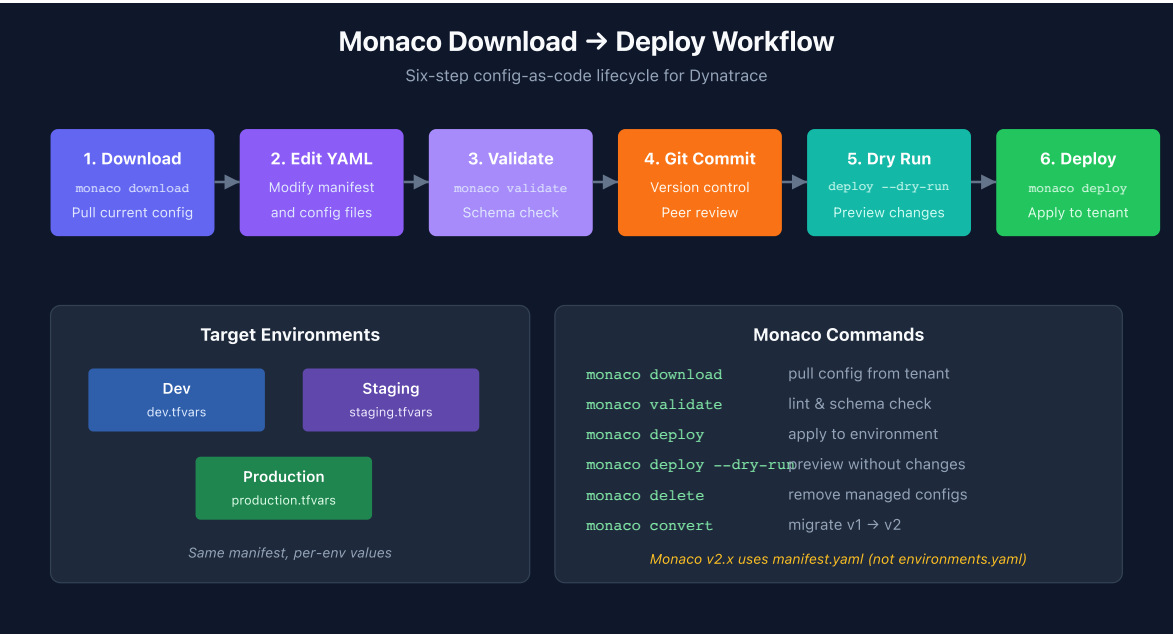
```
monaco delete manifest.yaml --file delete.yaml
```

**delete.yaml:**

```
delete:
  - type: builtin:management-zones
    id: old-management-zone-id
```

---

# Download vs Deploy Workflow



## 6. Advanced Features

### Configuration Dependencies

Reference other configurations:

```
configs:
  - id: my-alerting-profile
    config:
      name: Production Alerting
      template: alerting.json
      parameters:
        management_zone_id: "{{ .management-zones.production-mz.id }}"
    type:
      settings:
        schema: builtin:alerting.profile
        scope: environment
```

### Conditional Configurations

Skip configurations in certain environments:

```
configs:
  - id: production-only-config
    config:
      name: Production Monitor
      template: monitor.json
      skip:
        environments:
          - development
          - staging
    type:
      settings:
        schema: builtin:synthetic.http
        scope: environment
```

## Environment-Specific Overrides

```
configs:
  - id: alerting
    config:
      name: Alerting Profile
      template: alerting.json
      parameters:
        delay_minutes: 5
      overrides:
        - environments:
            - production
          override:
            parameters:
              delay_minutes: 1
    type:
      settings:
        schema: builtin:alerting.profile
        scope: environment
```

---

## Grouping Configurations

Organize related configs in groups:

```
configs:
  - id: web-app-configs
    group: web-application
    config:
      name: Web App Management Zone
      template: mz.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment

  - id: web-app-alerting
    group: web-application
    config:
      name: Web App Alerting
      template: alerting.json
    type:
      settings:
        schema: builtin:alerting.profile
        scope: environment
```

Deploy only a specific group:

```
monaco deploy manifest.yaml --group web-application
```

---

## Best Practices

| Practice                     | Description                                             |
|------------------------------|---------------------------------------------------------|
| <b>Use meaningful IDs</b>    | IDs should describe the config purpose                  |
| <b>Modular structure</b>     | Separate configs by domain (alerting, monitoring, etc.) |
| <b>Environment variables</b> | Never hardcode tokens or URLs                           |
| <b>Version control</b>       | Commit all changes with meaningful messages             |
| <b>Dry run first</b>         | Always validate before applying                         |
| <b>Document parameters</b>   | Comment complex configurations                          |

## Common Issues and Solutions

| Issue                   | Cause             | Solution                           |
|-------------------------|-------------------|------------------------------------|
| "Config already exists" | Duplicate ID      | Use unique IDs or download first   |
| "Schema not found"      | Typo in schema ID | Check exact schema name in API     |
| "Template not found"    | Wrong path        | Use relative path from config.yaml |
| "Validation failed"     | Invalid JSON      | Check JSON syntax in template      |

## 7. Next Steps

### When to Consider Terraform

| Scenario                   | Tool Choice |
|----------------------------|-------------|
| Config-only management     | Monaco      |
| Part of larger IaC stack   | Terraform   |
| Need state file management | Terraform   |
| Drift detection required   | Terraform   |
| Simple GitOps workflow     | Monaco      |

### Continue the Series

| Next Notebook                       | Focus                           |
|-------------------------------------|---------------------------------|
| <b>AUTOM-04: Terraform Provider</b> | Infrastructure-as-code approach |

### Additional Resources

- [Monaco Documentation](#)
- [Monaco Examples](#)
- [Configuration Schema Reference](#)

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## Summary

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In this notebook, you learned:

- How to install and configure Monaco
- Project structure and manifest configuration
- Creating and deploying configurations via YAML
- Advanced features like dependencies, overrides, and groups

**Key Takeaway:** Monaco is ideal for teams wanting GitOps-style configuration management. It provides the right balance of simplicity and power for most Dynatrace automation needs.

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*Continue to AUTOM-04: Terraform Provider to learn infrastructure-as-code patterns.*

## Community Resources & Examples

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The following GitHub repositories provide starter templates, real-world examples, and hands-on exercises for Monaco:

### Official Repositories

| Repository                                              | Description                                                                                                      |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <a href="#">dynatrace-configuration-as-code</a>         | Official Monaco CLI (v2.28.5) -- Go binary, Apache-2.0                                                           |
| <a href="#">dynatrace-configuration-as-code-samples</a> | Official samples repo with 9 Monaco starter templates in <code>basic-templates-monaco</code>                     |
| <a href="#">easytrade</a>                               | Demo microservices app with a working <code>monaco/</code> directory (manifest.yaml, detection rules, workflows) |

| Repository                               | Description                                                                                 |
|------------------------------------------|---------------------------------------------------------------------------------------------|
| <a href="#">Dynatrace-Config-Manager</a> | GUI tool for tenant-to-tenant config migration; complements Monaco for brownfield scenarios |

## Starter Templates (in `dynatrace-configuration-as-code-samples` )

| Template Directory                       | What It Configures                                                                                       |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>basic-templates-monaco</code>      | Alerting, app detection, synthetic, maintenance window, management zones, ownership, notifications, SLOs |
| <code>learn-monaco-auto-tag</code>       | Auto-tagging with Monaco                                                                                 |
| <code>account-monaco-admin-access</code> | Admin access setup for Monaco                                                                            |

## Pipeline Observability Samples

Monaco configurations for ingesting CI/CD pipeline events via OpenPipeline:

| Directory                                  | CI/CD Platform |
|--------------------------------------------|----------------|
| <code>github_pipeline_observability</code> | GitHub Actions |
| <code>gitlab_pipeline_observability</code> | GitLab CI      |
| <code>azure_devops_observability</code>    | Azure DevOps   |
| <code>argocd_observability</code>          | ArgoCD         |

## Training & Exercises

| Repository                                  | Description                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------|
| <a href="#">monaco-self-paced-exercises</a> | 6 structured exercises: install, auto-tag, download, variables, delete/restore, linking configs |
| <a href="#">monaco-demo</a>                 | Working GitHub Actions workflow for Monaco deploy with "crawl-walk-run" adoption methodology    |

**Note:** Monaco v1 ( `dynatrace-oss/dynatrace-monitoring-as-code` ) is deprecated.  
Use `monaco convert` to migrate v1 projects to v2.

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*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*